Networking

SONiC for OTN: Expanding SONiC and OCP SAI into Optical Transport Networks

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OTN Use Cases: DCI and AI Scale-Across

- **OTN DCI (Data Center Interconnection)**

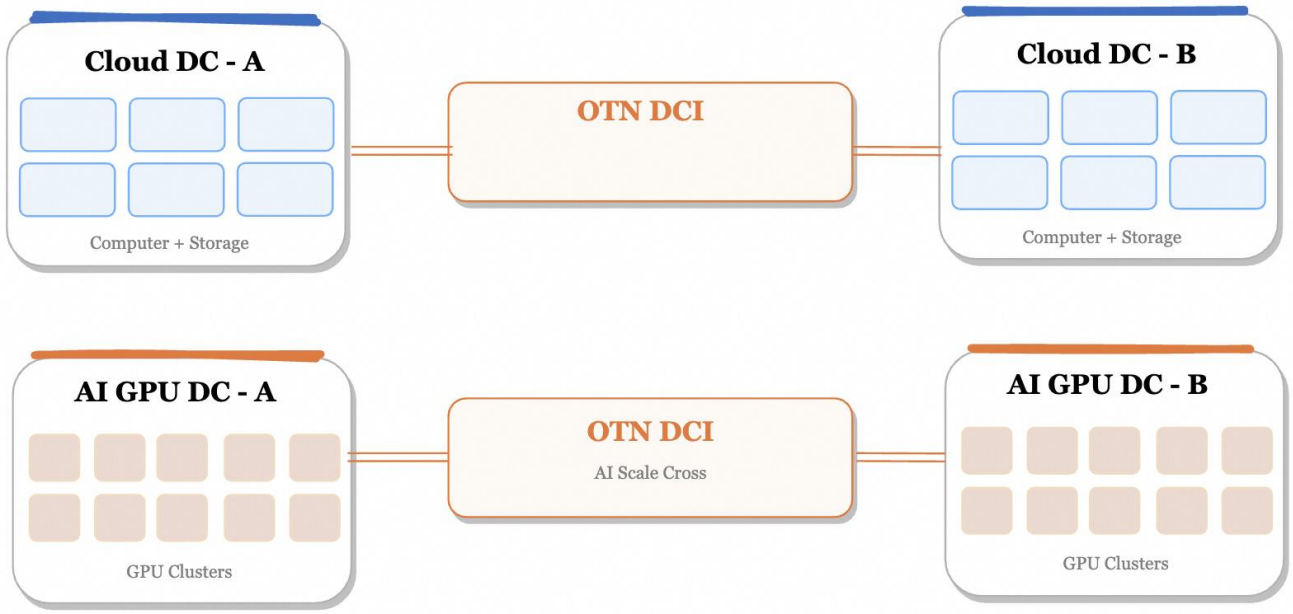
  Connecting datacenters in different regions and locations

- **AI Scale-Across**

  Multi-site GPU clusters operating as a single logical training system

- **Key requirements**

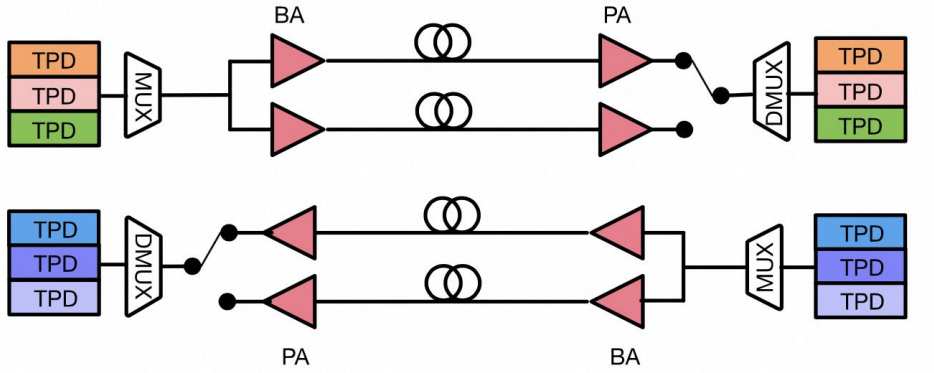
  High bandwidth, low latency, and high reliability



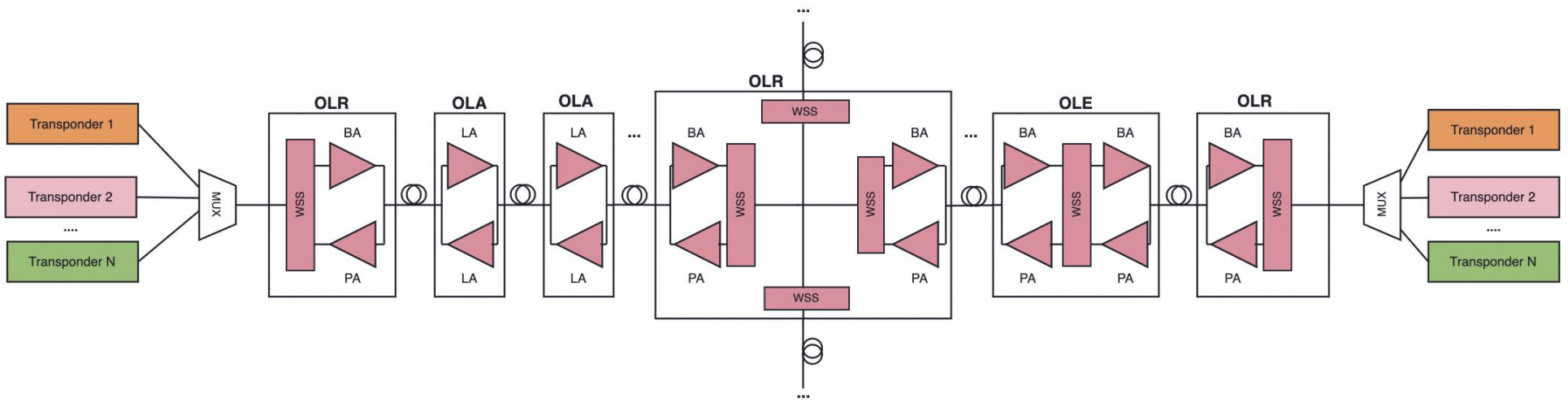
OTN Device Introduction

- A simple metro-DCI optical network with a point-to-point topology has the following optical devices

TPD : Transponder
PA : Pre-Amplifier
BA : Booster Amplifier
MUX : Wavelength Multiplexer
DMUX: Wavelength Demultiplexer
OLP : Optical Line Protection



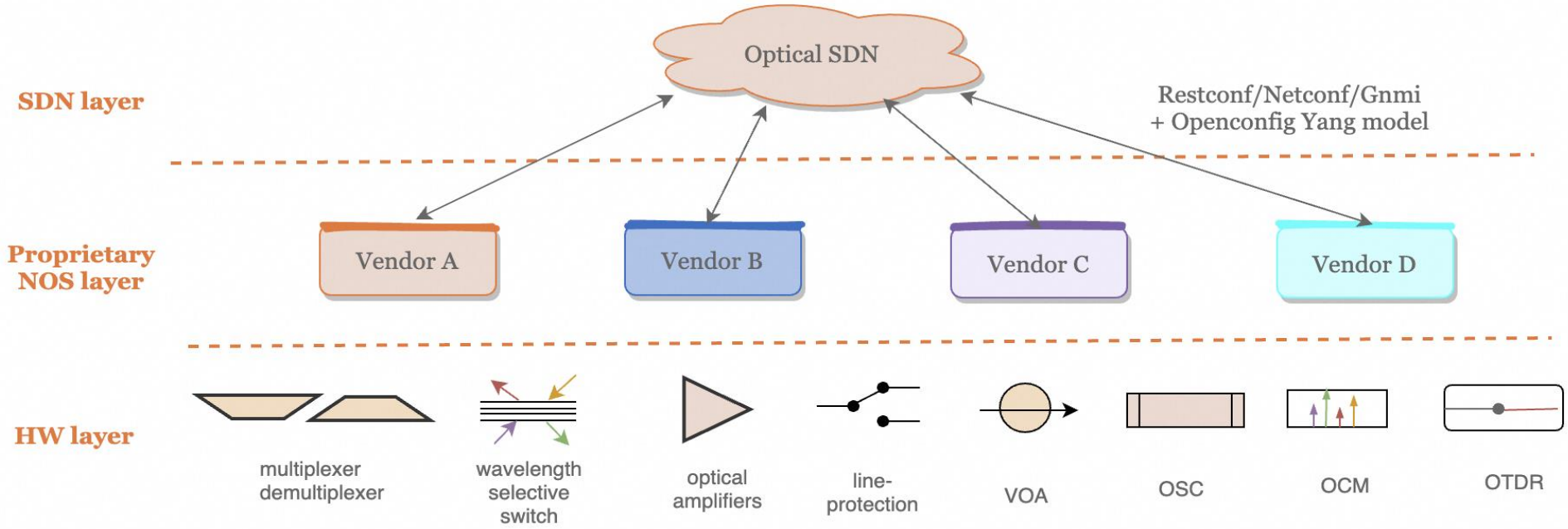
- A wide-area optical network with complex topology has many more optical devices



Challenges of Proprietary OTN NOS

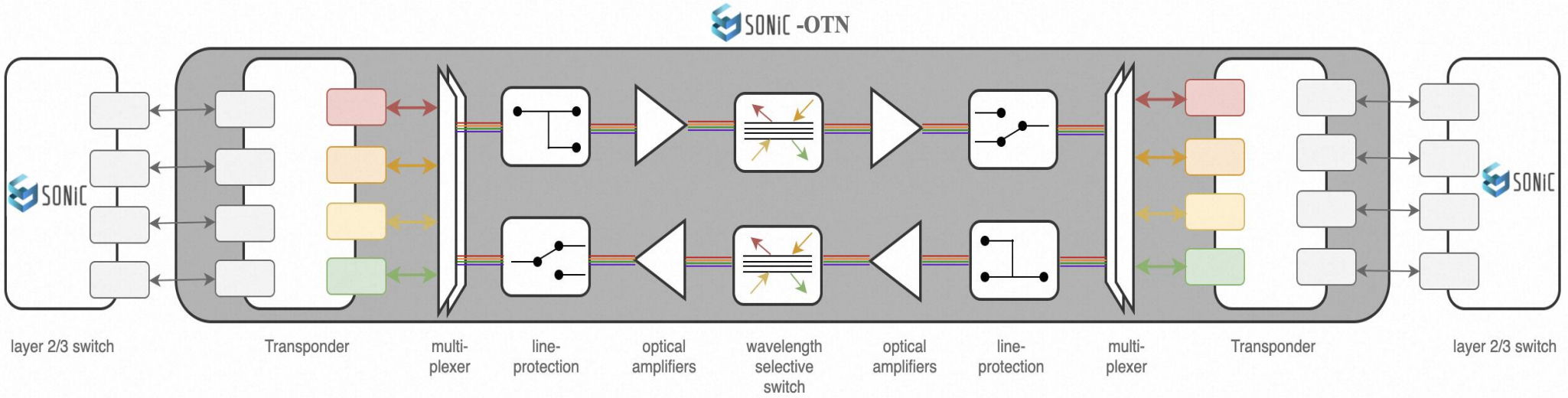
- **Gray-box OTN device**
  Proprietary NOS with standardized northbound interfaces and OpenConfig data model

  Fragmentation — Different CLI, alarms, and SDN adapters per vendor
  High OpEx — N vendors × N tool-chains × N training tracks
  Slow Patching — Every fix waits on the vendor's release schedule
  No Customization — Closed source; cannot adapt to AI-agent development loops



SONiC For OTN Working Group

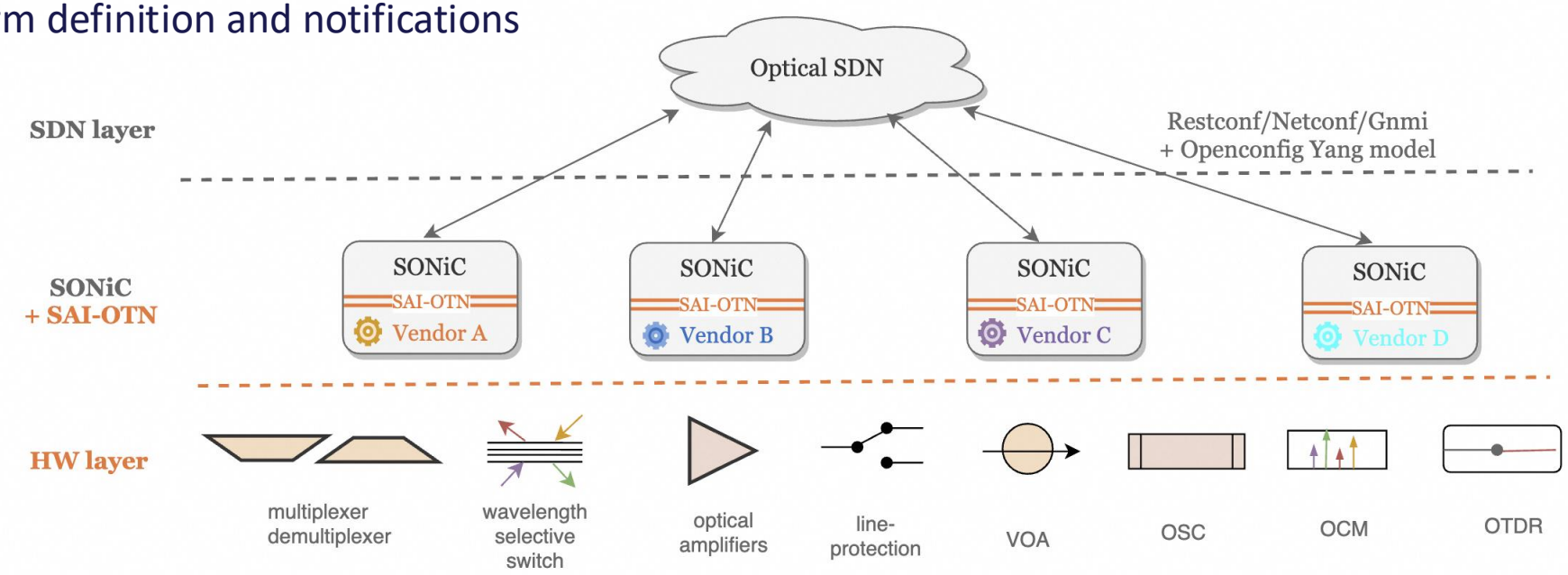
- SONiC OTN working group was established in January 2023
- Goal: Enable SONiC support for optical transport equipment (transponders, line systems, etc.)
- Contributors: Alibaba, Microsoft, Molex, Nokia, Cisco, Accelink
- Homepage: https://lists.sonicfoundation.dev/g/sonic-wg-otn
- GitHub repo: https://github.com/sonic-otn



What SONiC + SAI Brings to OTN

- White-box devices with SONiC NOS, sharing the SONiC ecosystem and community
  Unified device northbound interfaces (Restconf, GNMI, SNMP, CLI, etc.)
  Unified system management (user management, security, in service upgrading, …)
  Open-source codebase, AI-agents can accelerate the development–testing–operating loop

- OCP SAI with OTN, sharing OCP SAI infrastructure and community
  Unified CRUD APIs for all optical components
  Unified alarm definition and notifications



Value for Operators and Vendors

- **As an OTN network operator**

  No Vendor Lock-in, Lower Cost

  - Separates software from hardware — free to choose any white-box OTN device
  - Drive down equipment costs with multi-vendor

  Fully Programmable, Full Control

  - Customize network behavior without waiting for vendor updates
  - Build your own features for AI networking and AI-agent integration

  One OS for Everything, Easy to Scale

  - Same system across all switches, routers and OTN devices
  - Manage thousands of OTN devices with automation

- **As an OTN device vendor**

  Faster Time to Market

  - Build on a mature open-source base with closer customer collaboration
  - Focus resources on hardware innovation and differentiation

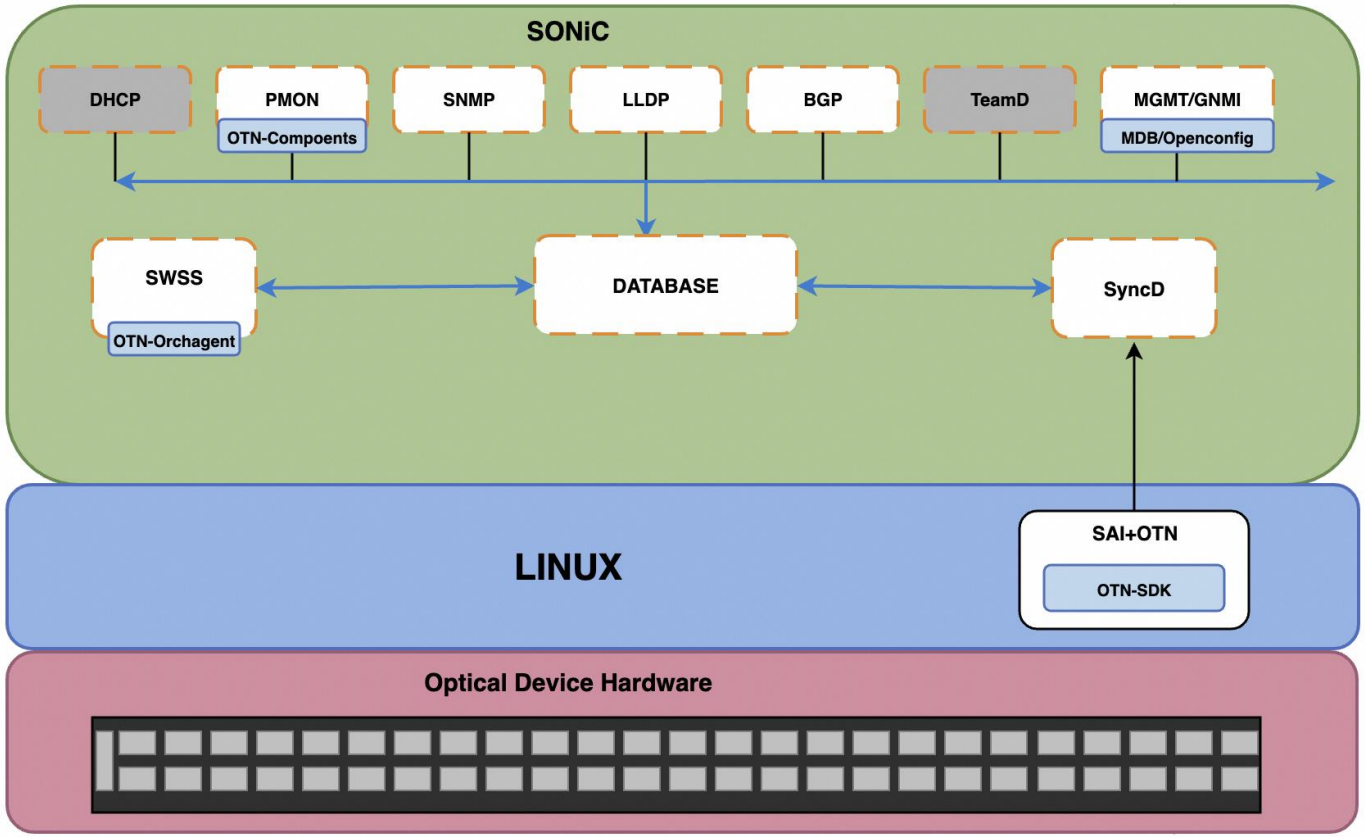
  Shared Development, Lower R&D Cost

  - Leverage community contributions — bug fixes, new features, testing at scale
  - Reduce long-term software maintenance burden

SONiC for OTN Architecture

- **Extends SONiC with OTN**
  - Reuses SONiC architecture — no reinvention
  - Full consistency with SONiC
  - Extends to optical transport domain

- **Modules Extended for OTN**
  - **SAI** — SAI+OTN extension for optical components
  - **Syncd** — OTN SAI objects monitoring
  - **SWSS** — OTN SAI objects orchestration agent (orchagent) and management
  - **PMON** — Optical hardware components monitoring
  - **MGMT** — Multi-ASIC support and OpenConfig YANG models

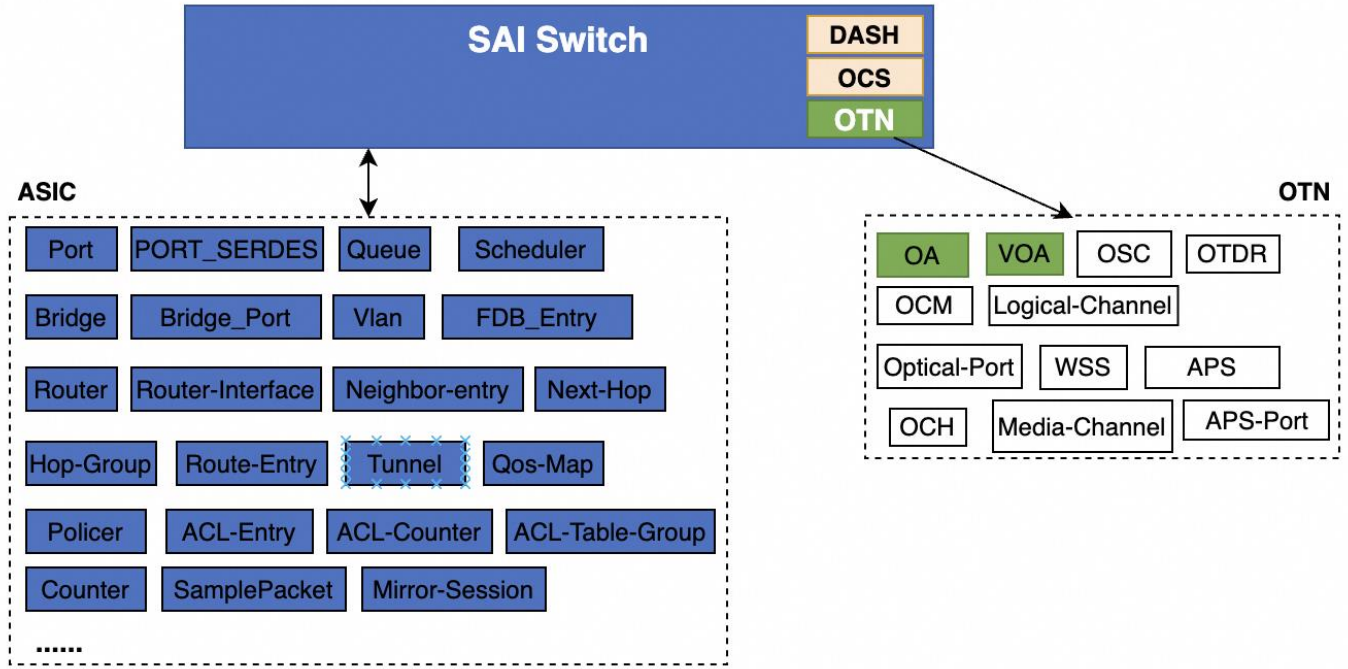


OCP SAI — Expanding to Optical

- Switch Abstraction Interface (SAI): A standardized, vendor-agnostic hardware abstraction API

- **SAI Scope — Expanding Beyond Packet ASIC**
  - Packet Switch ASIC — original scope, L2/L3 forwarding
  - DASH — SmartNIC / DPU dataplane abstraction
  - OCS — Optical Circuit Switch
  - **OTN — Optical Transport Network**

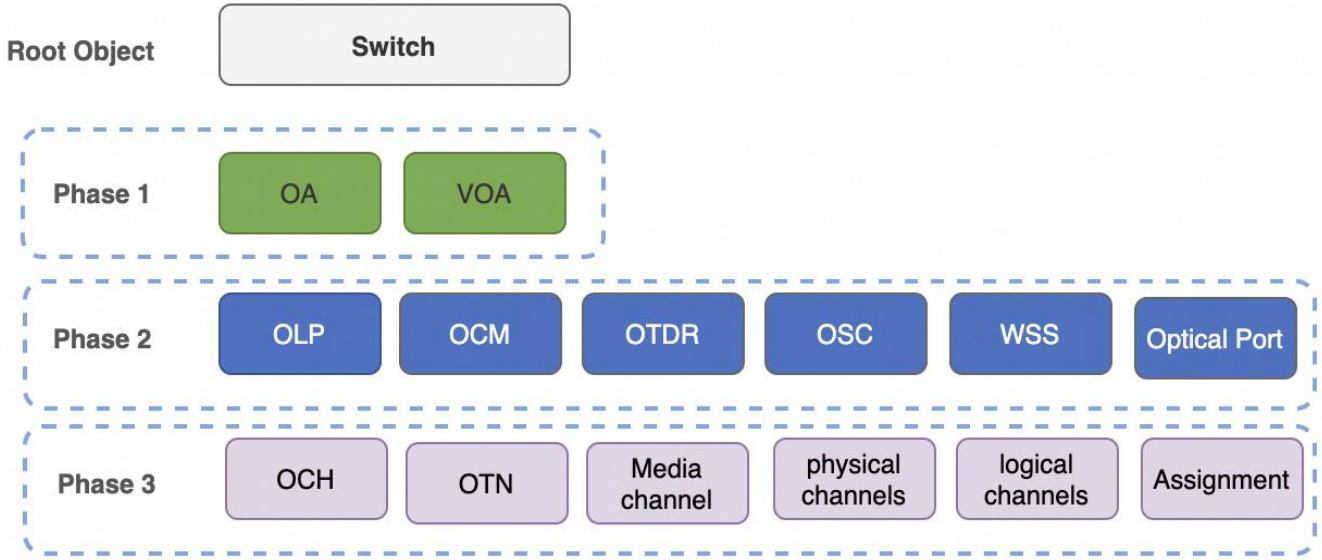
- **SAI + OTN**
  - Consistent with DASH and OCS project
  - Designated as SAI experimental objects
  - Based on OpenConfig YANG models
  - Zero overlap or interaction with existing IP/packet SAI objects



SAI OTN Implementation Phases

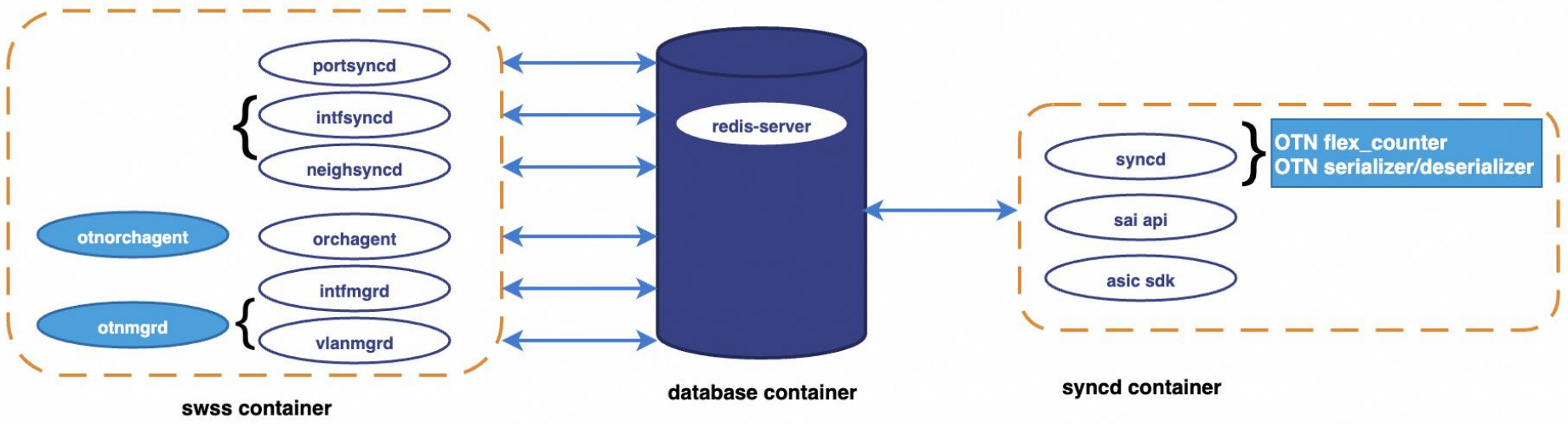
- All OTN object — sub-objects of the SAI switch root
- Standard CRUD APIs per object
  - create / remove / set_attribute / get_attribute
- New @precision tag
  - enables floating-point values, such as optical power, Bit Error Rate(BER)

- Phase 1 (Delivered):
  - OA (Optical Amplifier), VOA (Variable Optical Attenuator)
  - minimum viable optical device support
  - PR #2217 merged
- Phase 2 (In Progress):
  - Full Open Line System
  - OCM, OTDR, WSS, APS, OSC, Optical Port
- Phase 3 (Planned):
  - Transponder support
  - Logical Channels, Optical Channels (OCH, OMC), etc.



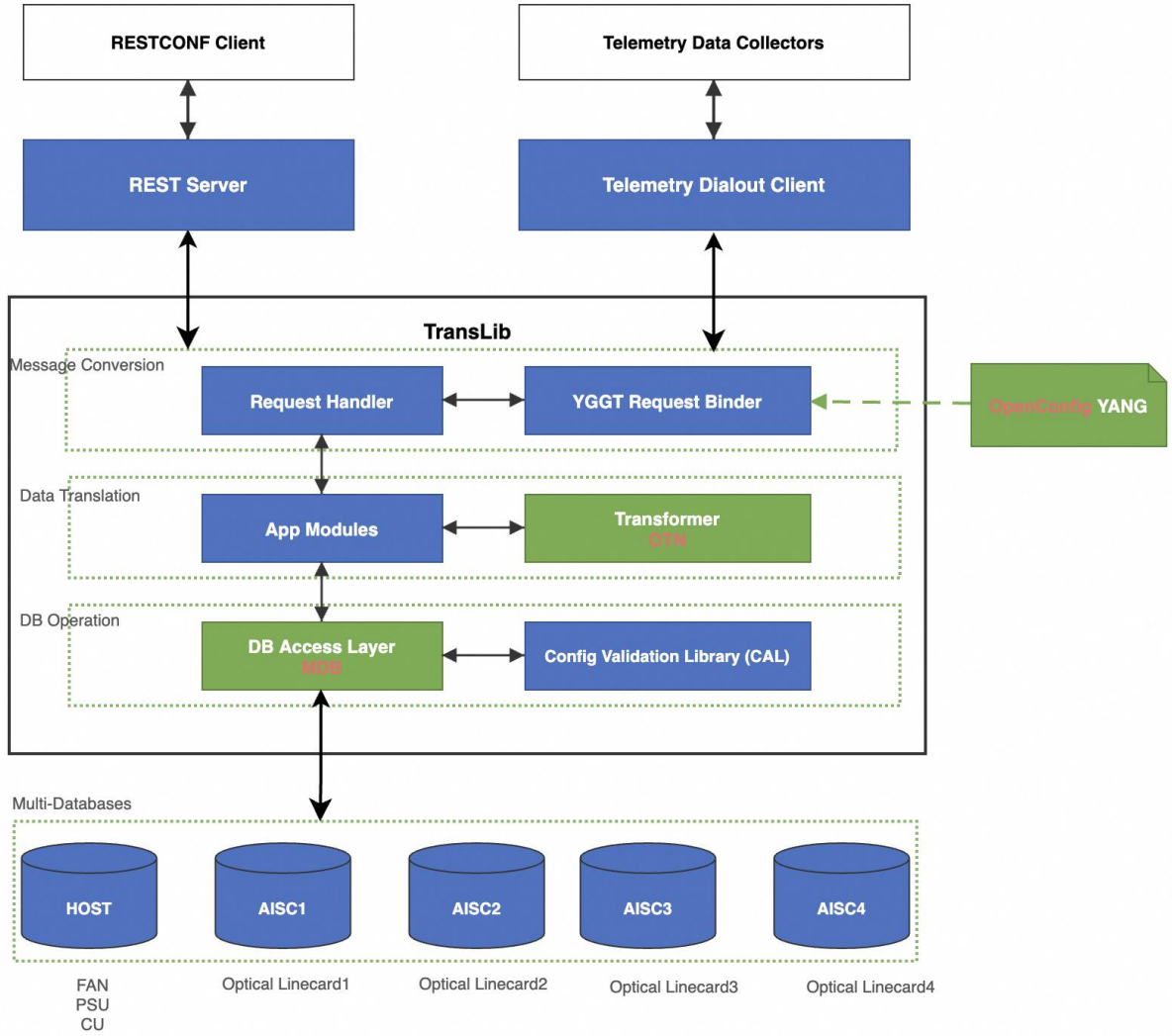
SWSS and Syncd modules for OTN

- Syncd: synchronizes hardware state with the Redis database via SAI
  - Add FlexCounter for SAI OTN objects, and OTN metadata serializer/deserializer
  - PR in progress, PR 2 (https://github.com/sonic-otn/sonic-sairedis/pull/2)
- SWSS: manages network orchestration and state synchronization
  - Add OTN management and orchestration agent
  - PR in progress, PR 3 (https://github.com/sonic-otn/sonic-swss/pull/3)



Multi-ASIC and Northbound APIs for OTN

- SONiC supports multi-ASIC architecture
  - Pizza-Box OTN device: uses a single ASIC
  - Multi-slot OTN device: each optical linecard maps to an ASIC namespace
- Extend Mgmt-framework and GNMI module
  - Supports multi-database (MDB)
  - Supports OpenConfig YANG models.
    - openconfig-optical-amplifier.yang
    - openconfig-optical-attenuator.yang
- HLD and all code PR are in progress
  - HLD PR #1776: https://github.com/sonic-net/SONiC/pull/1776
  - MDB PR #14: https://github.com/sonic-otn/sonic-mgmt-common/pull/14
  - OA/VOA PR #15: https://github.com/sonic-otn/sonic-mgmt-common/pull/15
  - WSS, OCM, OTDR, OPS PR #16



SONiC OTN KVM — Try It

- SONiC OTN KVM image
  - Virtual machine enabling testing without physical OTN hardware.
  - Download pre-built image: https://github.com/sonic-otn/sonic-buildimage/releases
  - KVM Instructions: https://github.com/sonic-otn/sonic-buildimage/blob/otn-dev/platform/otn-kvm/README.md
  - KVM Northbound Interface Example: https://github.com/sonic-otn/sonic-buildimage/blob/otn-dev/platform/otn-kvm/OTN-KVM-NBI-Example.md

```
admin@sonic:~$ show platform firmware status
Chassis     Module        Component       Version  Description
----------  ------------  ------------  ---------  ------------------------------------------------------------
OLA         LINE-CARD0    OA0-0                 1  Optical amplifier (west)
                          OA0-1                 1  Optical amplifier (east)
                          OSC0-0                1  Optical supervisory channel (west)
                          OSC0-1                1  Optical supervisory channel (east)
                          OCM0-0                1  Optical channel monitor
                          OTDR0-0               1  Optical time domain reflectometer
            SUPERVISOR0   BIOS                  1  Performs initialization of hardware components during booting
                          FPGA                  1  Platform managment controller for on-board components
                          CPLD                  1  Used for managing IO modules
                          ONIE                  1  Open network install environment
admin@sonic:~$
```

```
admin@sonic:~$   curl -k -X GET\
   "https://127.0.0.1/restconf/data/openconfig-optical-amplifier:optical-amplifier/amplifiers" \
   -H "accept: application/yang-data+json" | jq
```

```
curl -k  -X PUT \
   -H "Content-Type: application/yang-data+json" \
   -H "Accept: application/yang-data+json" \
   "https://127.0.0.1/restconf/data/openconfig-optical-amplifier:optical-amplifier/amplifiers/amplifier=OA0-0/confi
   -d '{"openconfig-optical-amplifier:target-gain": "5.0"}' | jq
```

```
gnmic -a 127.0.0.1:8080 -u admin -p YourPaSsWoRd --insecure \
   get --path openconfig-optical-attenuator:optical-attenuator/attenuators
```

Summary

- **OTN demand:**
  - Introduced the OTN for DCI and AI scale-across;
  - Discussed gray-box devices, proprietary NOS, and associated challenges

- **Solution:**
  - OCP SAI + OTN and SONiC OTN workgroup
  - Why use OTN white-box device with SAI and SONiC?
  - SONiC-OTN workgroup: scope, goals and members

- **Progress:**
  - Extend SAI and SONiC with OTN, the architecture
  - Optical objects and features in SAI, and current status
  - The OTN enhancement of SWSS/Syncd, northbound interfaces, and in-progress PRs

- **Try it:**
  - How to build and run SONiC-OTN KVM
  - Test the CLI, Restconf, and GNMI

Call to Action

- **Join the SONiC for OTN working group**
  - Working group homepage: https://lists.sonicfoundation.dev/g/sonic-wg-otn
  - Project mailing list: sonic-wg-otn@lists.sonicfoundation.dev
  - Bi-weekly meeting: Tuesdays 6:00 PM, PDT

- **Contribute to SAI OTN Development**
  - Review and comment on SAI OTN experimental objects (Phase 2 & 3)
  - GitHub: https://github.com/sonic-otn

- **OTN Device Vendors: Adapt Your Hardware to SONiC-OTN**
  - Implement SAI-OTN vendor adapter for your optical devices
  - Contribute code to SONiC OTN modules (SWSS, Syncd, Mgmt, etc)

- **Get Started Today**
  - Download SONiC OTN KVM image and try it now: https://github.com/sonic-otn/sonic-buildimage/releases

Thank You!

